

Machine Learning: Unit 2 Decision Trees and Ensembles

VTU 2022 Scheme CBCS - BCS602 Module 2

--- MODULE OVERVIEW & LEARNING OBJECTIVES ---

1. Decision Tree Learning: Entropy, Information Gain, and Gini Impurity calculations
2. ID3 Algorithm: Step-by-step splitting on categorical attributes with numerical example
3. Overfitting in Decision Trees: Pre-pruning and Post-pruning strategies
4. Ensemble Learning: Bagging vs Boosting vs Stacking
5. Random Forests and AdaBoost: Weighted voting and misclassification penalties

--- IMPORTANT VTU 10-MARK QUESTIONS ---

- Q1. Calculate Entropy and Information Gain for given training table [10 Marks]
- Q2. Explain AdaBoost training algorithm with neat diagram [10 Marks]